

Lab 8

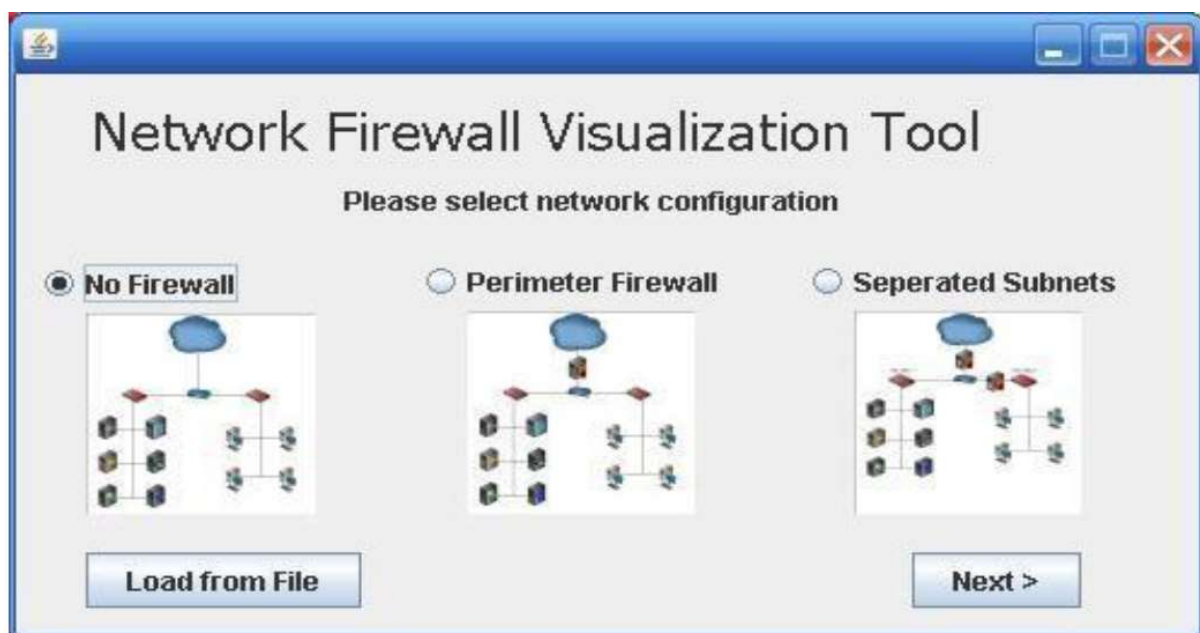
Lab requires installation of a firewall visualization tool (written in Java) downloaded from KevinCurran's website (prof at University of Ulster). This requires java. His site contains many other useful and relevant resources and it would be beneficial to include them in the references.

Firewall Virtualization Tool Introduction

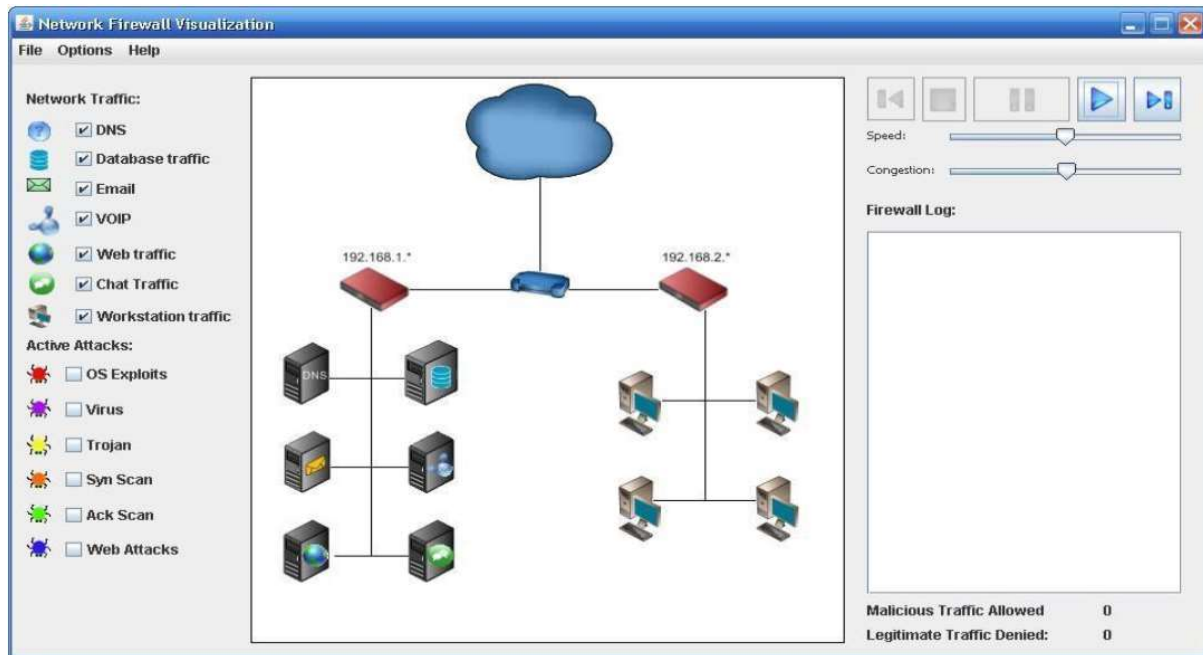
1. Start the Firewall program by downloading the visual tool.

- <https://kevincurran.org/com320/labs/Firewall/FirewallVisualizationTool.jar>
- <https://kevincurran.org/com320/labs/Firewall/FirewallWorkstationDataFile.dat>

2. You should see a screen similar to the one below:



Choose "**no firewall**" and click next. The following screen will appear:

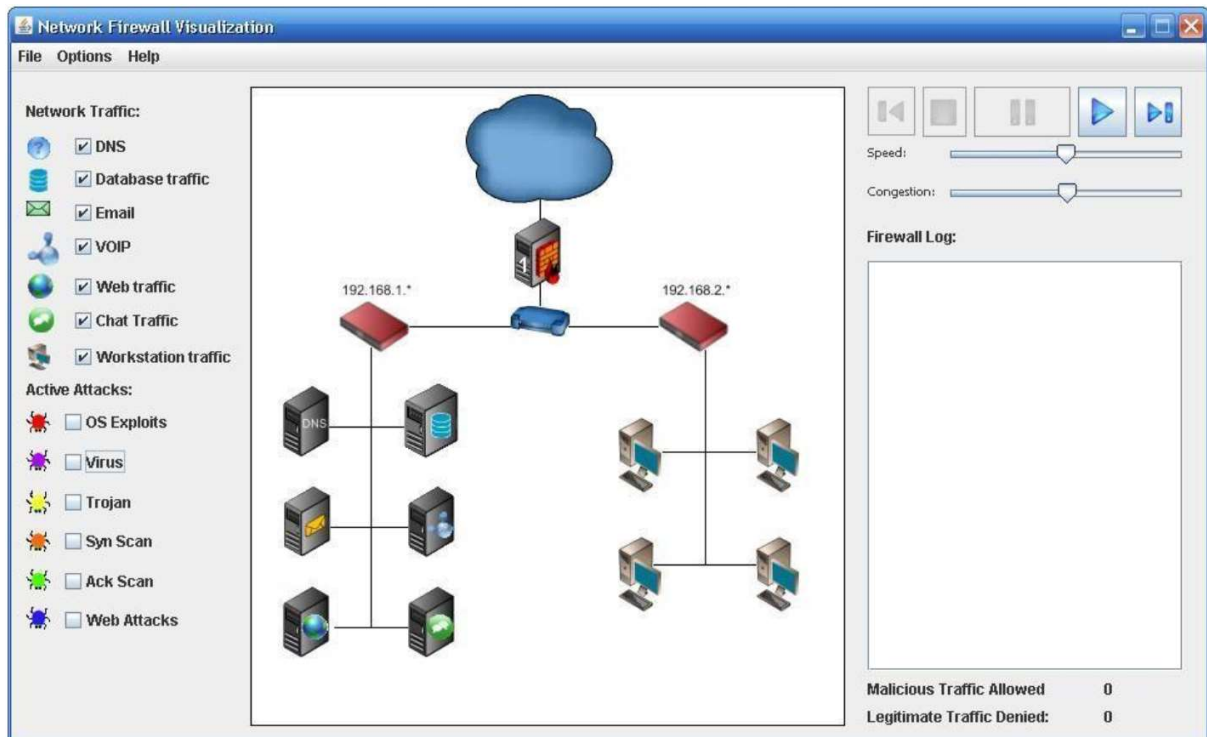


Click  the button. Note that the traffic flows both from the “cloud” or internet to the client machines. By default, there is no malicious traffic flowing to the machines. Click on the *OS Exploit* option. Eventually, you’ll see a similar red coloured bug flow from the internet into the local area network and land on a machine, infecting the machine. Once a machine is infected, it is marked as such with the “international No” emblem. Let us see how configuring a firewall will help prevent such infections.

Firewall Configuration.

1. Start a new session by clicking **File -> New** in the upper window of the tool. This time, choose the

Perimeter firewall. The window that comes up will look like this:

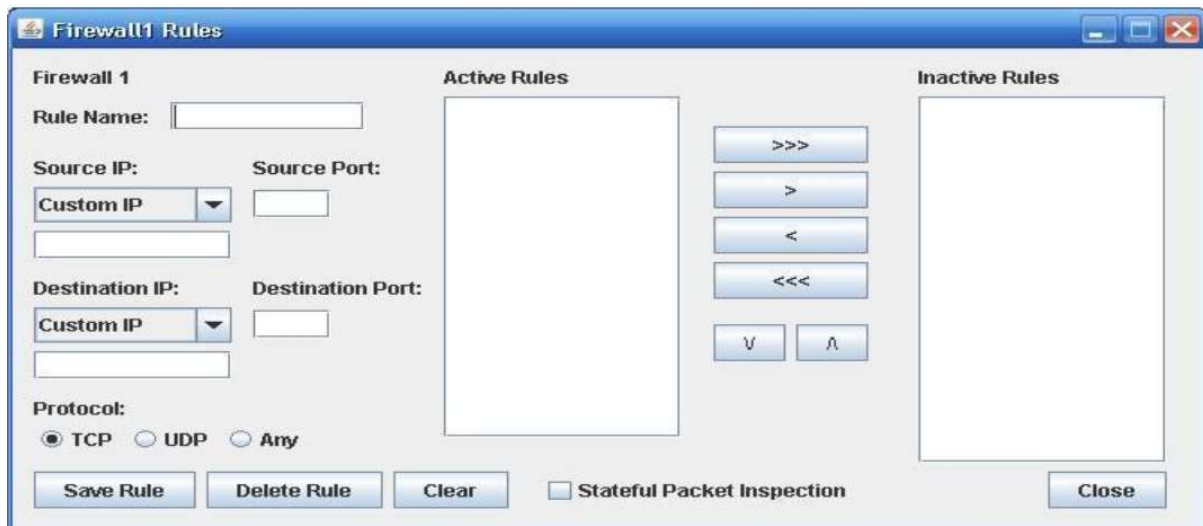


You now have a firewall between the internet (represented by a cloud) and your network router. Click  the play button and watch what happens. **Do you see traffic flowing from the internet into your system or from your network to the internet? Explain why or why not:**

2. Add some active attacks by clicking on several different options.

3. Configure your firewall to allow traffic to flow in and out of your network. Do this by choosing

the 'options' tab at the top of tool & define firewall rules. You should see a screen like below:



Name your firewall rule (typically with a name that focuses on a given subject or attack). The “Source IP” option and port refer to how you want the firewall to recognize a given source IP/Port combination and respond. The Destination is similar but focusing on a destination rule. The goal of any good firewall configuration is to identify legitimate traffic while restricting malicious traffic. Try setting the following firewall rule:

Rule Name: DNS Rule

Source IP: DNS, Source Port: 53

Destination IP: Any, Destination port * Protocol: Any.

Click “**Save Rule**”. You should now see the rule in your Active Rules box. Click “close” and you should be back to your Network Firewall Visualization Tool window. Click the play button and watch what happens. You may need to move the speed bar to the right for a higher speed of traffic. Add some active attacks and watch if they flow through the firewall.

4. Download the **WorkstationDatabase** scenario from the above. It is called **FirewallWorkstationDataFile.dat1** and save it to your desktop. Choose **File -> new** to restart the program and click “**load from file**” button, pointing the program to the file you downloaded.

<https://kevincurran.org/com320/labs/Firewall/FirewallWorkstationDataFile.dat>

This scenario was configured so that workstations can pass through *firewall2* and gain access to the database. *Firewall1* has an **allow all** traffic rule set so all information is passed through to the network and from the network to the servers. Write rules to prevent active attacks from passing through *firewall 1* and attacking the database.

5. Would you claim your rule is now sufficient to allow traffic to flow for a typical network? Why or why not?

6. Do any of the active attacks now work against machines behind the firewall?
7. Create a rule to permit Email traffic to enter the network and check which of the active attack now work against machines behind the firewall?